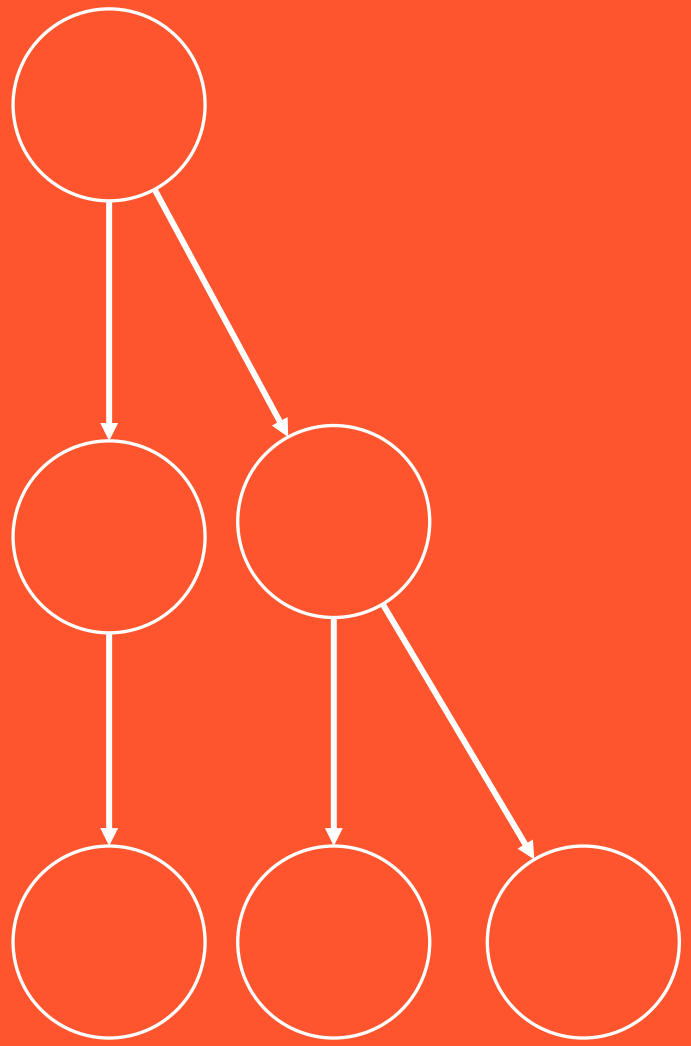


BCOG 100

Spatial Cognition

Lecture 9.2 — Keeping Track, Making Maps, and Exploring

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Out and Back, Blindfolded

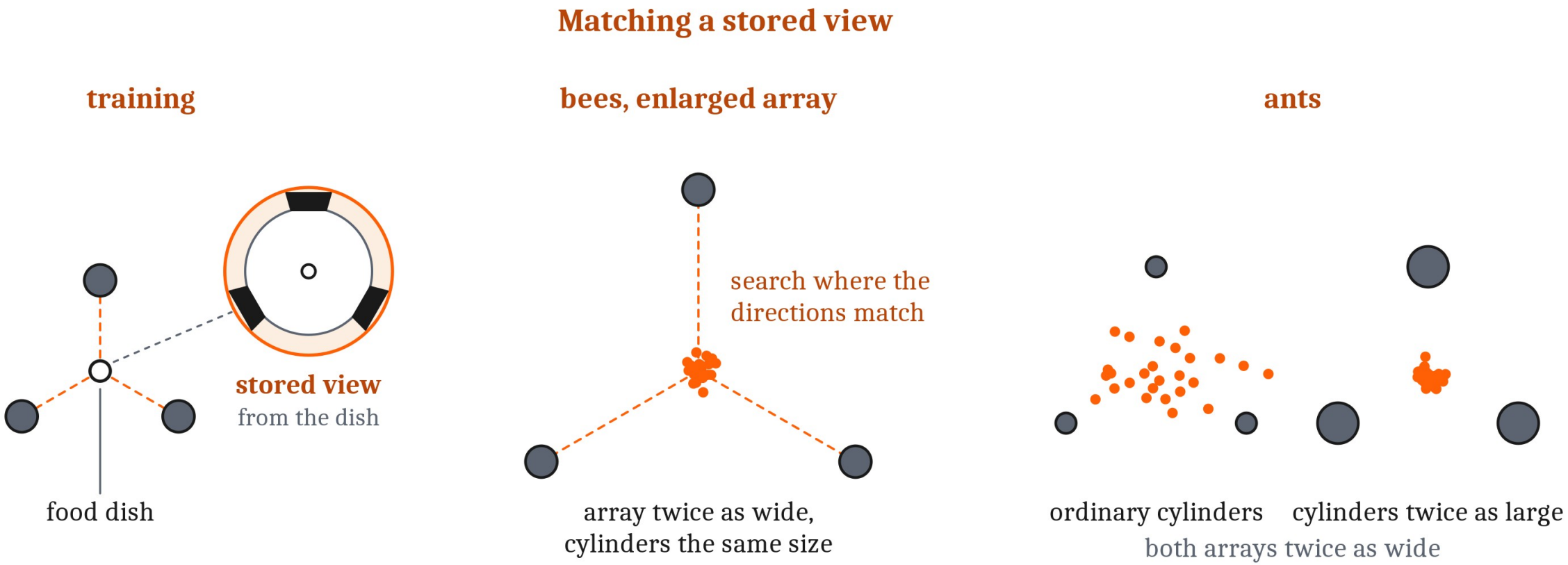
In your group, three minutes.

Imagine walking with your eyes closed: twenty steps ahead, a turn to the left, ten more steps. Then you have to point back to where you started.

How would you keep track? What would go wrong, and what would help?

Beacons, Snapshots, and Routes

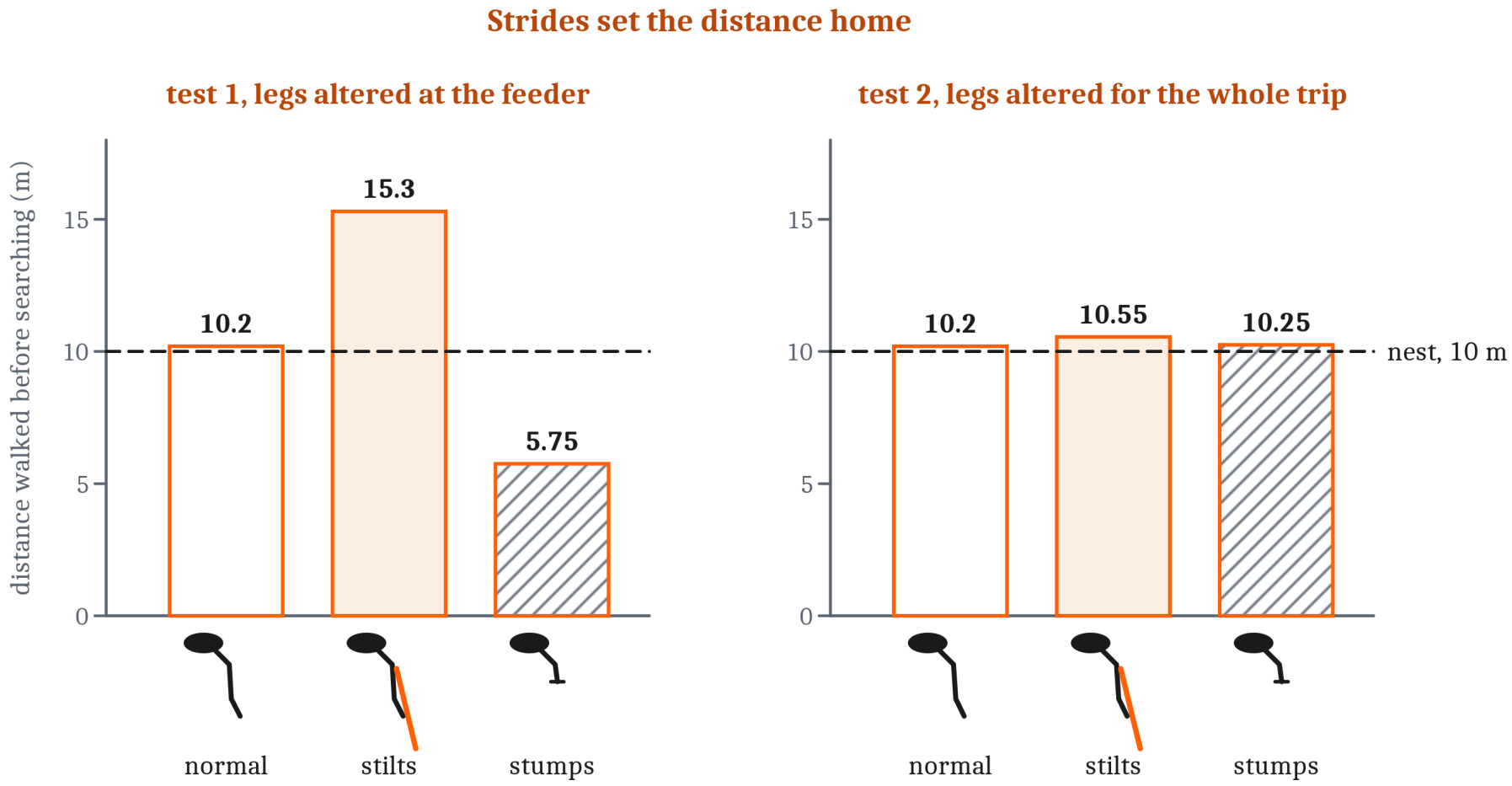
A **snapshot** is a stored view of the surroundings from the goal. A **route** is a fixed sequence of moves, or of landmarks, leading from one place to another.



from above; dots mark where the insects searched; schematic, after Cheng and Graham (2013)

A Running Sum

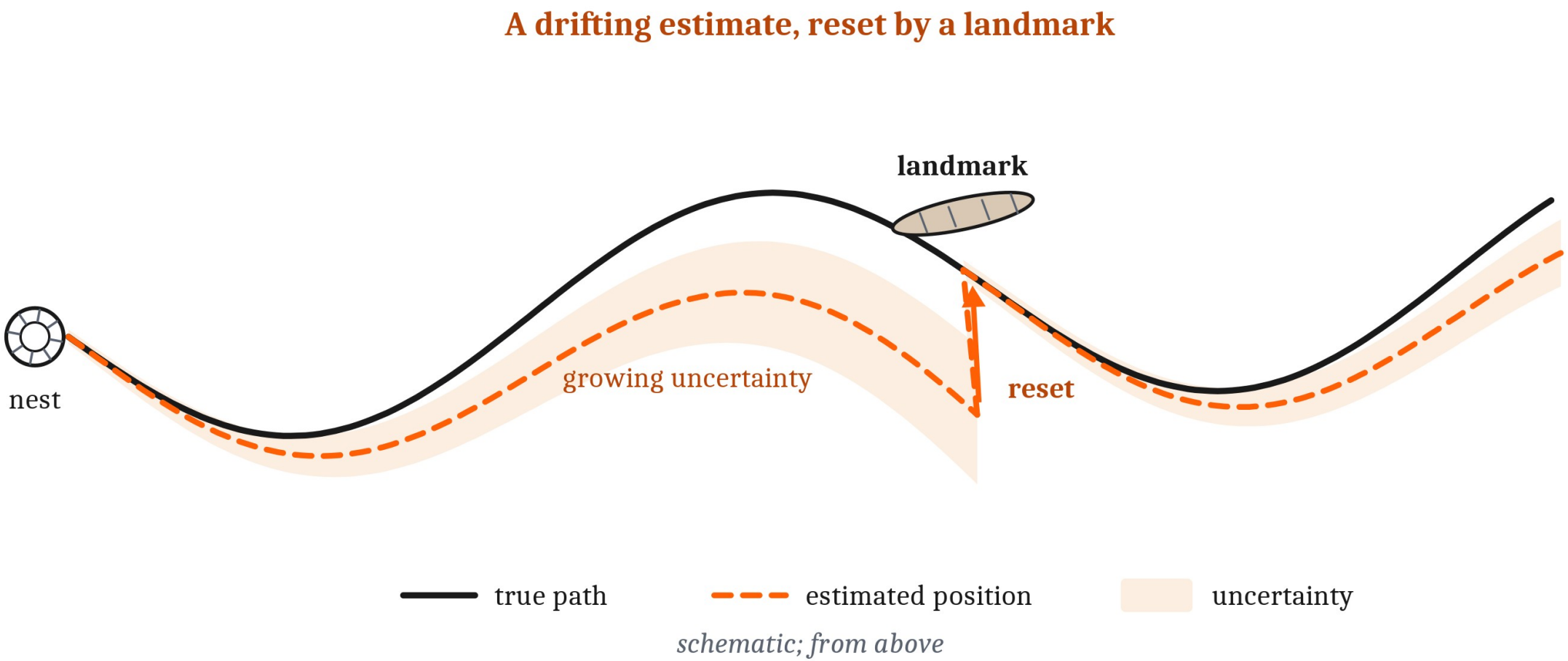
Path integration: a running estimate of the straight-line distance and direction back to the start, from heading and distance traveled. The result is a **home vector**.



medians, 25 ants per group; redrawn from the printed values of Wittlinger, Wehner, and Wolf (2006)

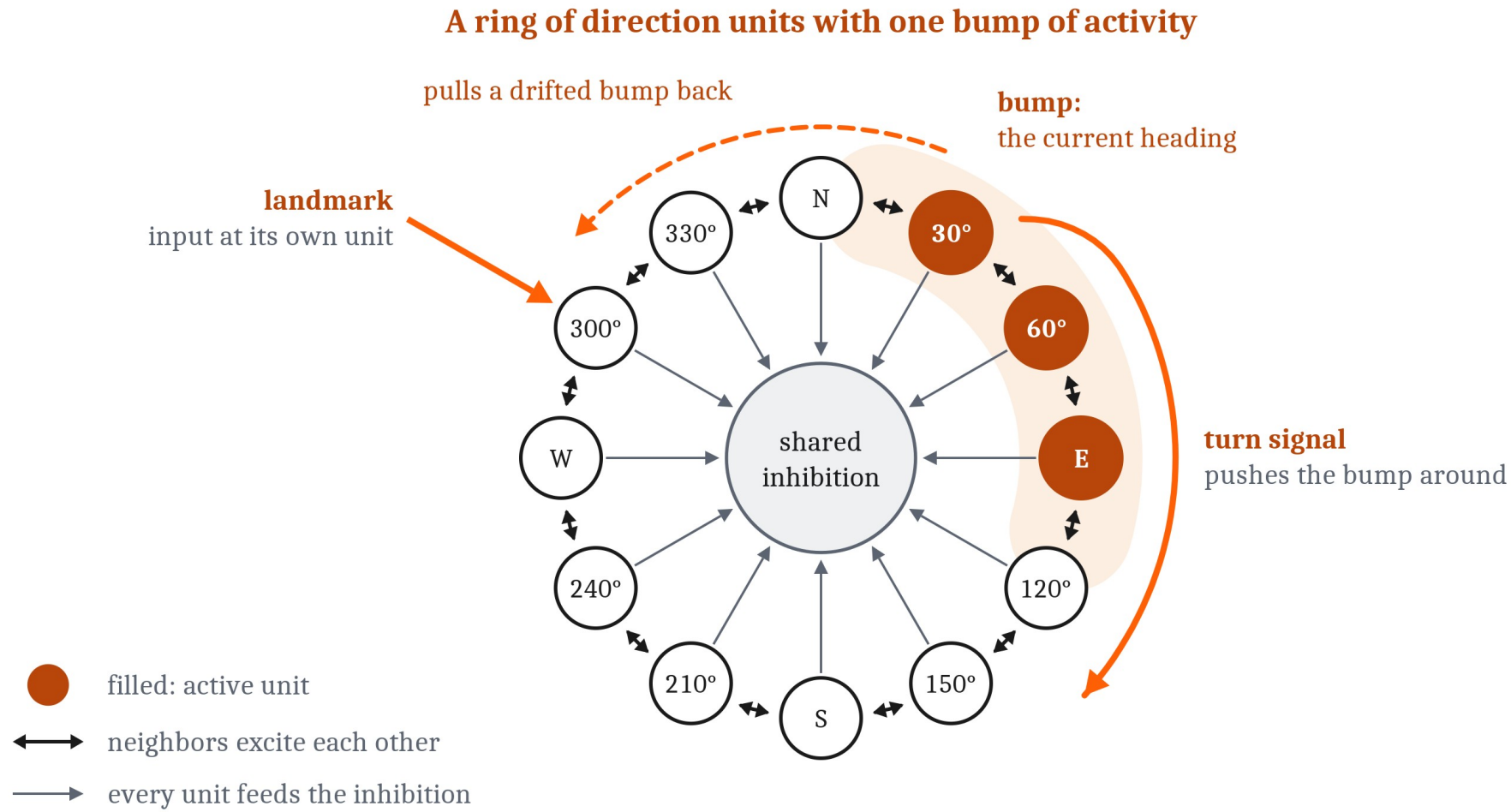
Correcting the Sum

Three corrections: averaging by **cue combination**, combining estimates each weighted by how reliable it is; resetting; and following one source. A landmark is worth using only if it stays put.



Keeping a Heading

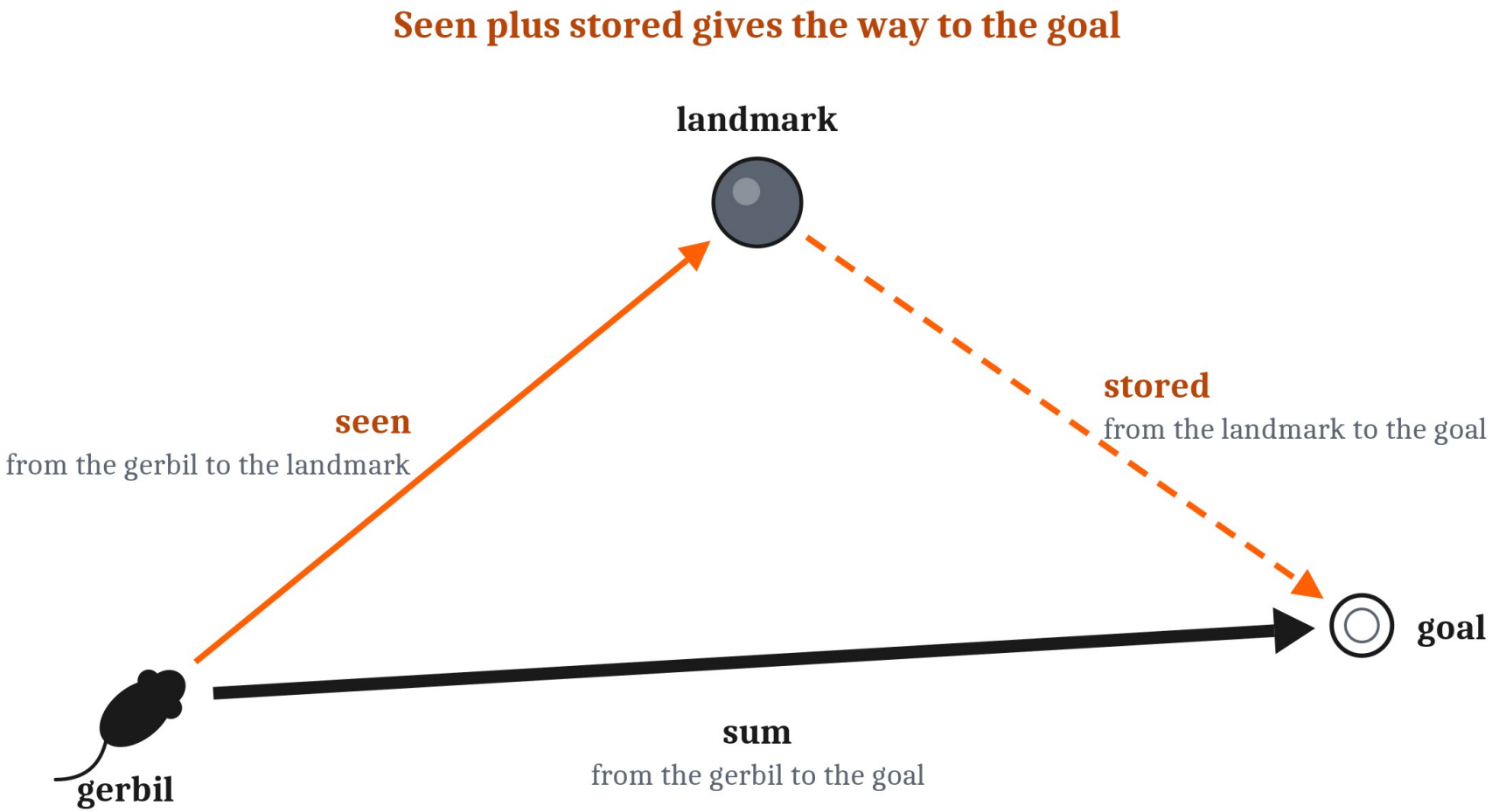
A **ring attractor**: units on a circle, each standing for one direction, that excite their neighbors and share inhibition. One bump of activity marks the heading.



our own schematic

From the Animal's Directions to the World's

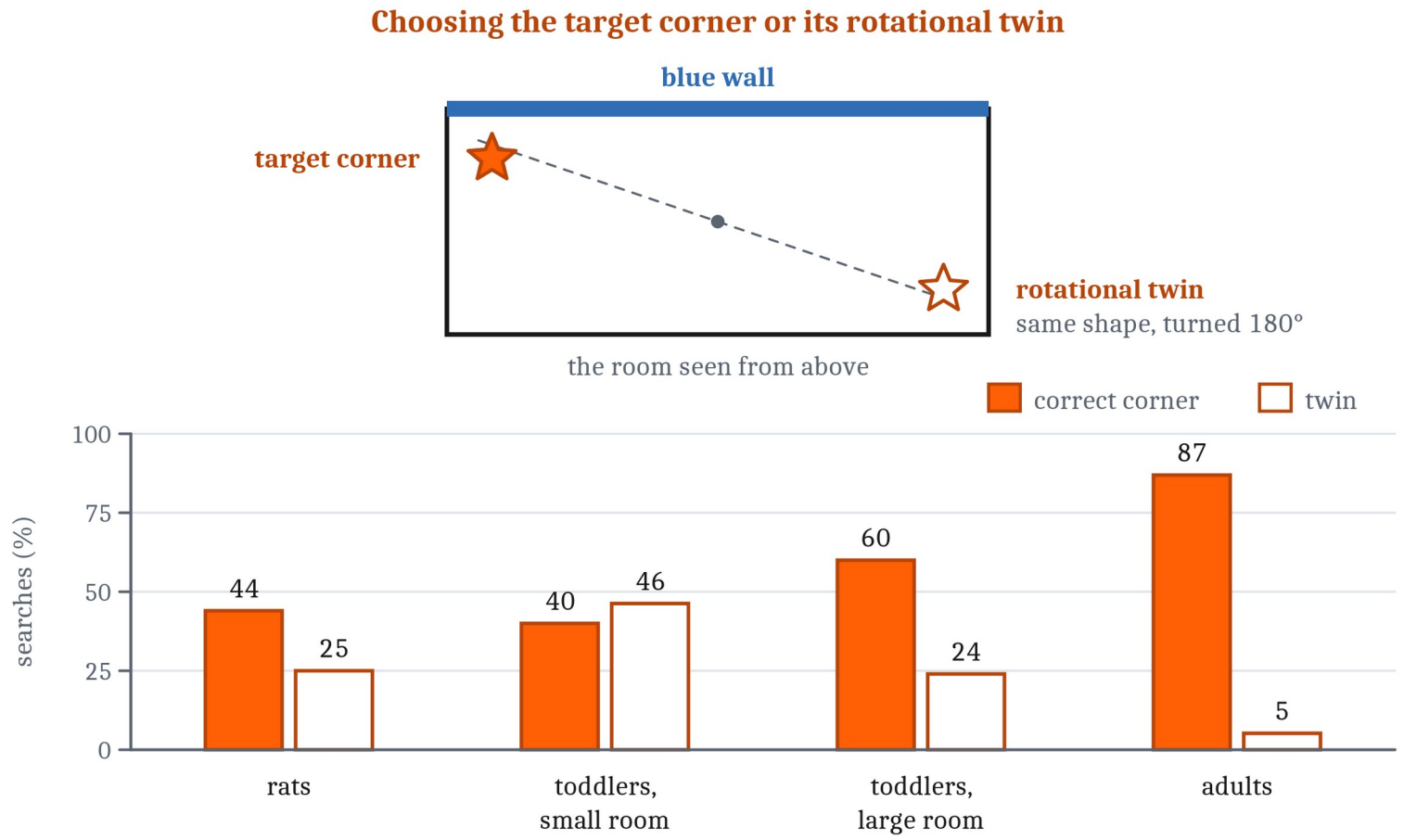
An **egocentric** location is relative to the animal; an **allocentric** one, to things in the world. Each is stated in a **frame of reference**. Converting between them is vector addition.



schematic, from above; after Cheng and Graham (2013)

Reorienting by Shape

Reorientation is recovering which way one is facing, and where one is, after becoming disoriented. Which cue an animal relies on depends on how reliable and noticeable it has proved.

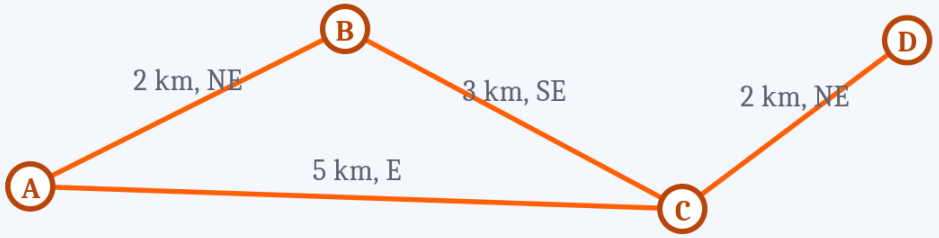


percentages from Cheng and Newcombe (2005); rats, Cheng (1986), with a distinct panel in each corner; toddlers and adults with a blue wall; large room, Learmonth and colleagues (2001); adults, Hermer-Vazquez and colleagues (1999)

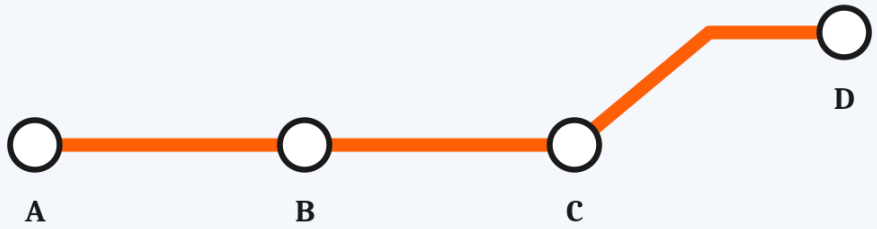
What a Map Contains

Survey knowledge: where places lie relative to one another. A map contains it; Tolman's **cognitive map** is one. A **metric map** has distances and directions; a **topological map**, only connections.

A metric map, like a sailor's chart



A topological map, like a subway map



which places connect to which, and nothing more

Repeating codes, combined (spacings invented for the drawing)



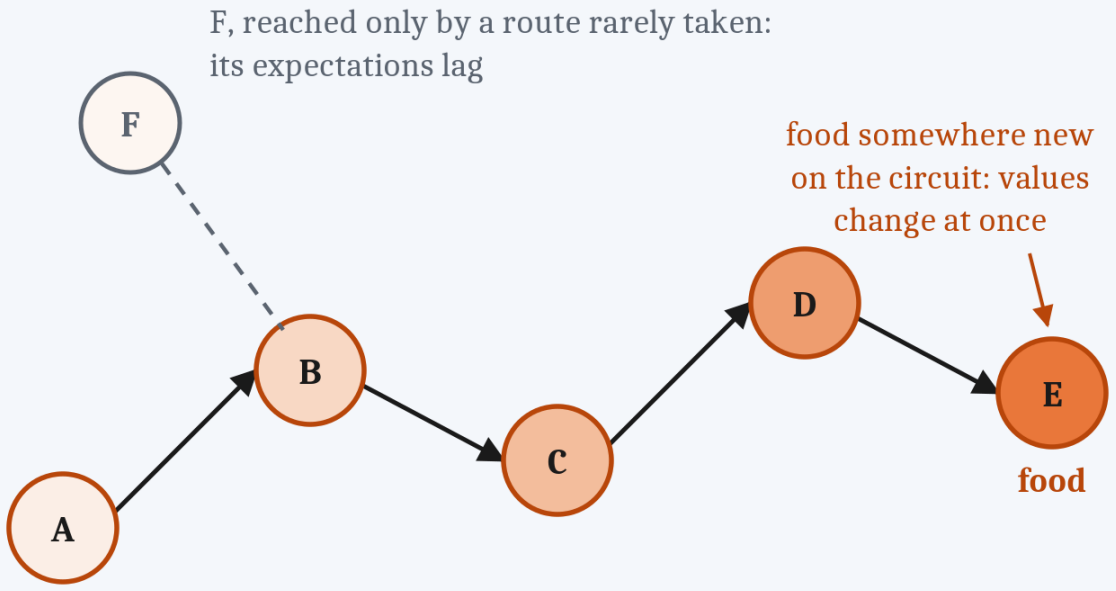
the one place where all three line up: the animal is here

Each code alone fixes position only within one repeat; together they pin down one place.

Using the Map, and Choosing Which One

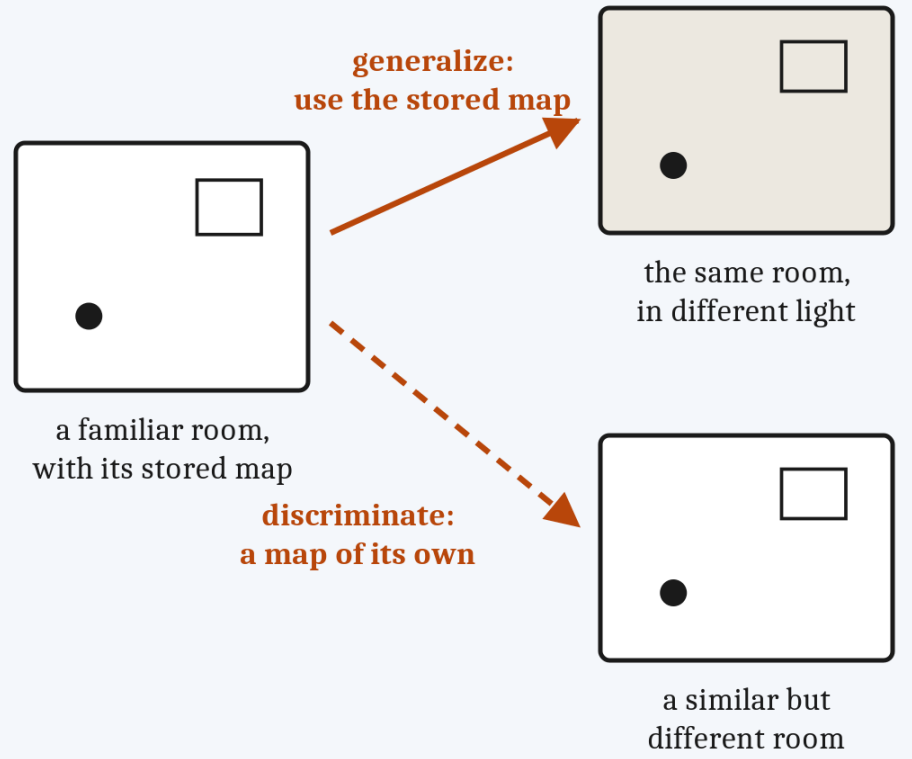
The map is the model in model-based choice. One proposal, a *predictive map*, stores for each place an expectation of which places will follow it, and how soon. And an animal has to choose which map it is on.

A predictive map



Usual circuit A to E. Each place's value comes from which places follow it, and how soon.

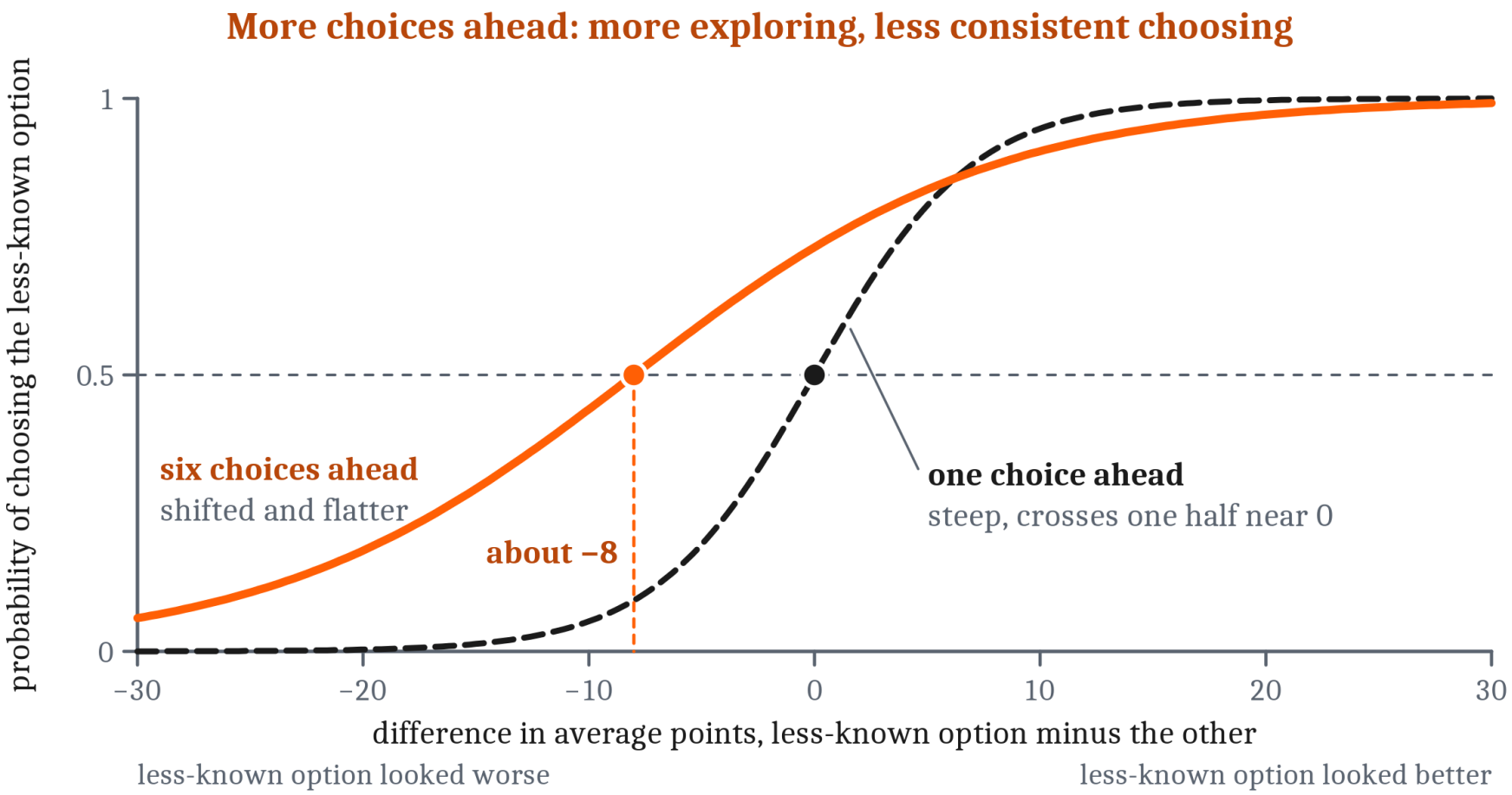
Which map am I on?



The predictive map is quick and only partly flexible; and every place needs the right map.

Ways to Explore

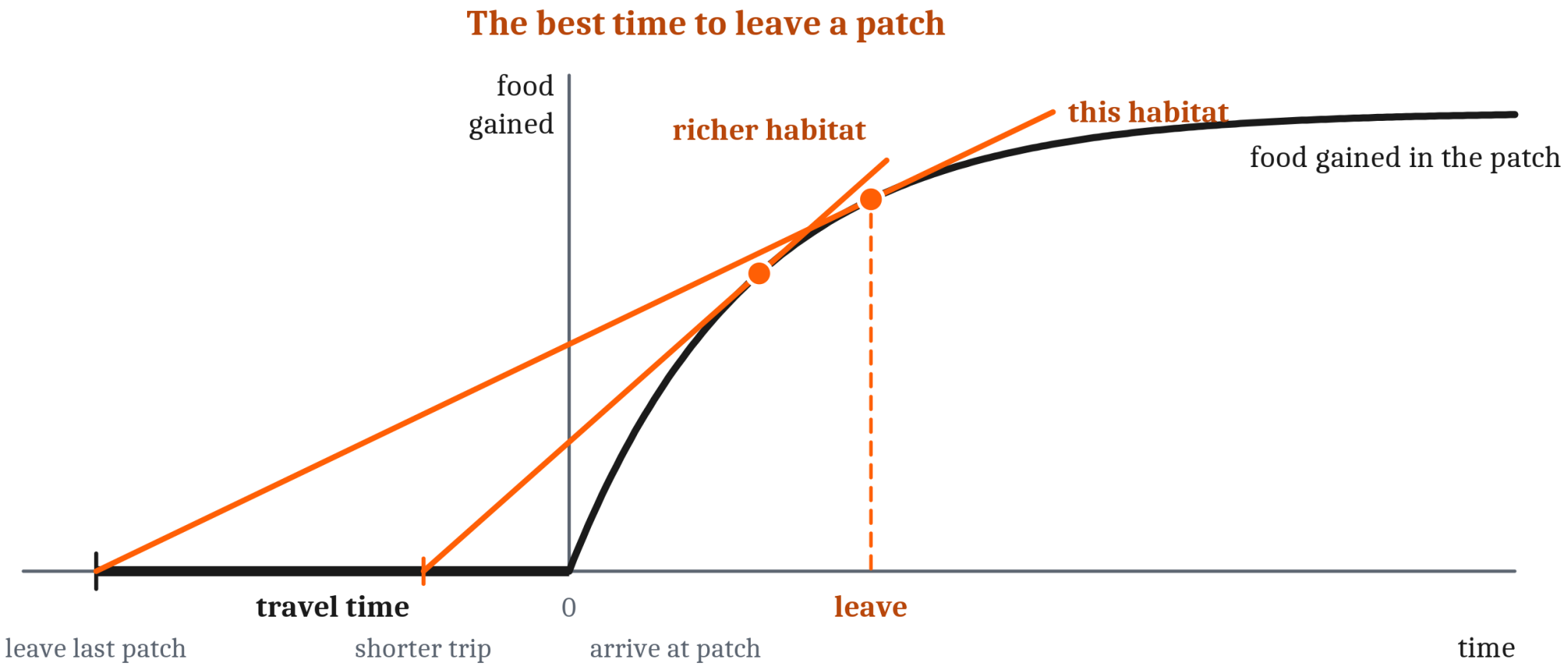
Random exploration: mostly the better-looking option, sometimes another. **Directed exploration**: a bonus for less-known options. A **novelty bonus**: extra value for being new.



schematic curves redrawn after Wilson and colleagues (2014); the −8 crossing as reported, slopes not fitted

Ways to Explore

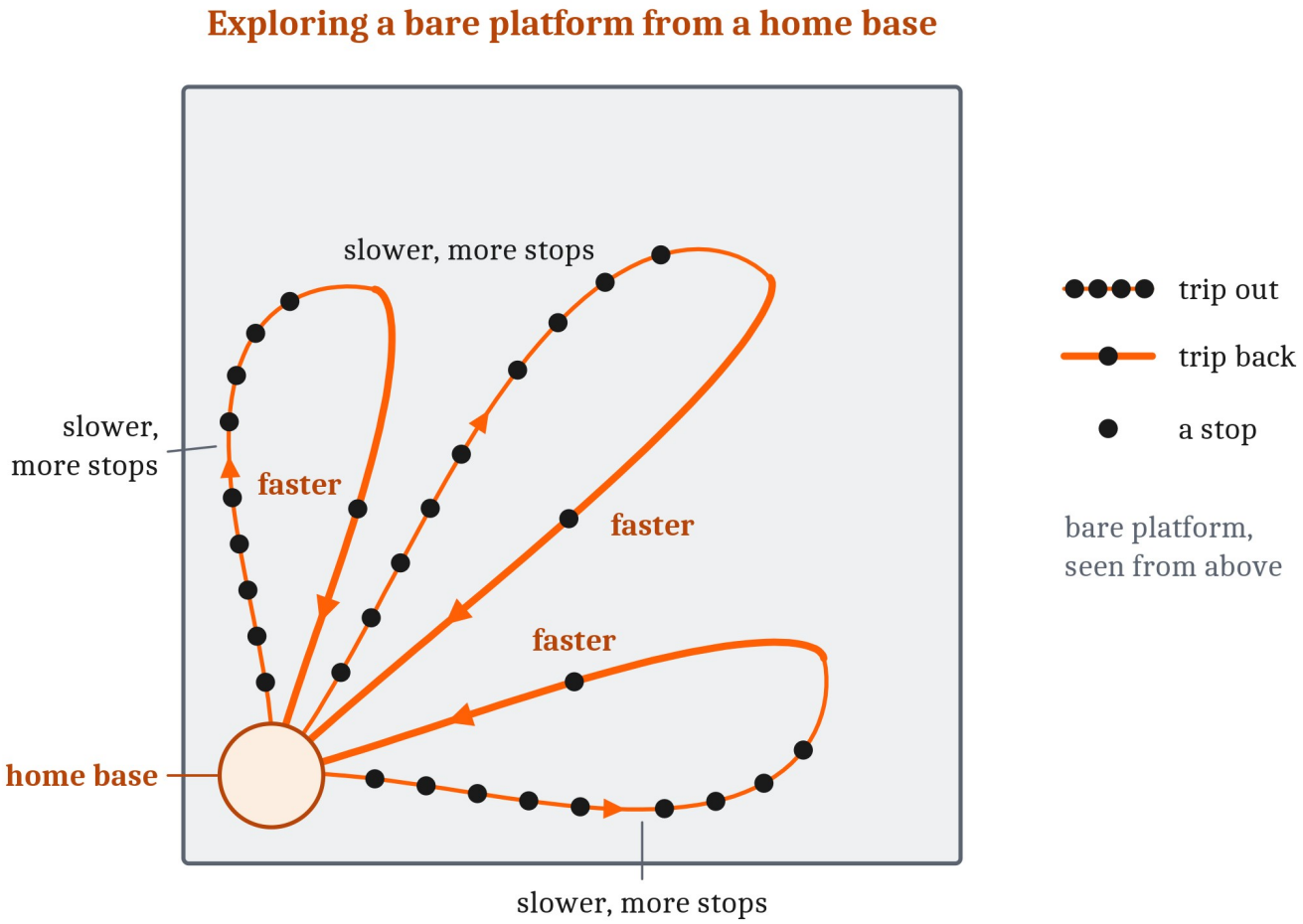
Random exploration: mostly the better-looking option, sometimes another. **Directed exploration**: a bonus for less-known options. A **novelty bonus**: extra value for being new.



our own drawing of Charnov's (1976) rule; the slope of each line is the average rate of food over trip and patch

Ways to Explore

Random exploration: mostly the better-looking option, sometimes another. **Directed exploration**: a bonus for less-known options. A **novelty bonus**: extra value for being new.

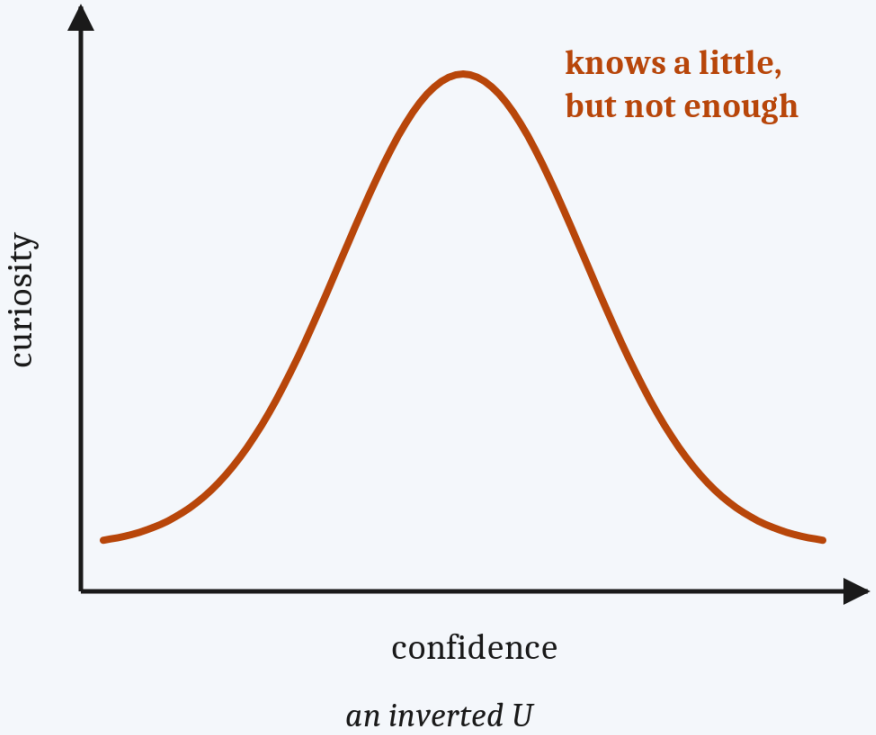


schematic after Eilam and Golani (1989); path shapes and stop counts are not drawn as measured

Curiosity: Information, or Progress?

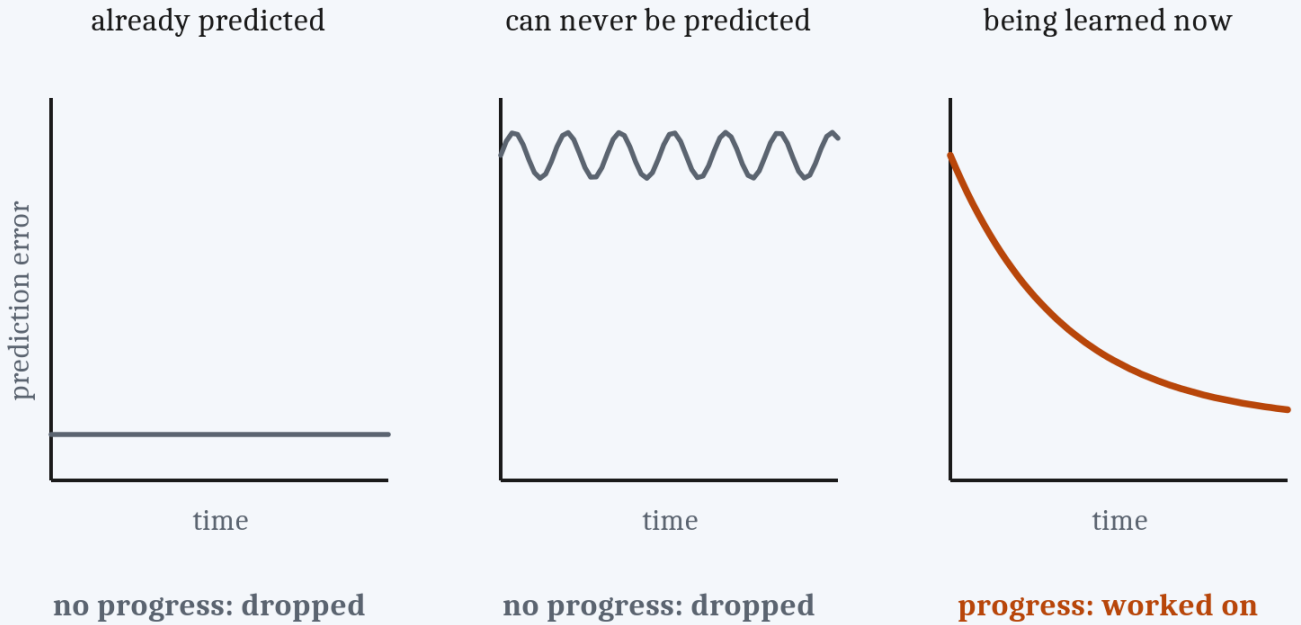
Curiosity is a drive to gain information for its own sake. Two proposed answers to what it rewards: information itself, or progress in predicting.

Information as the reward



an inverted U

Progress as the reward



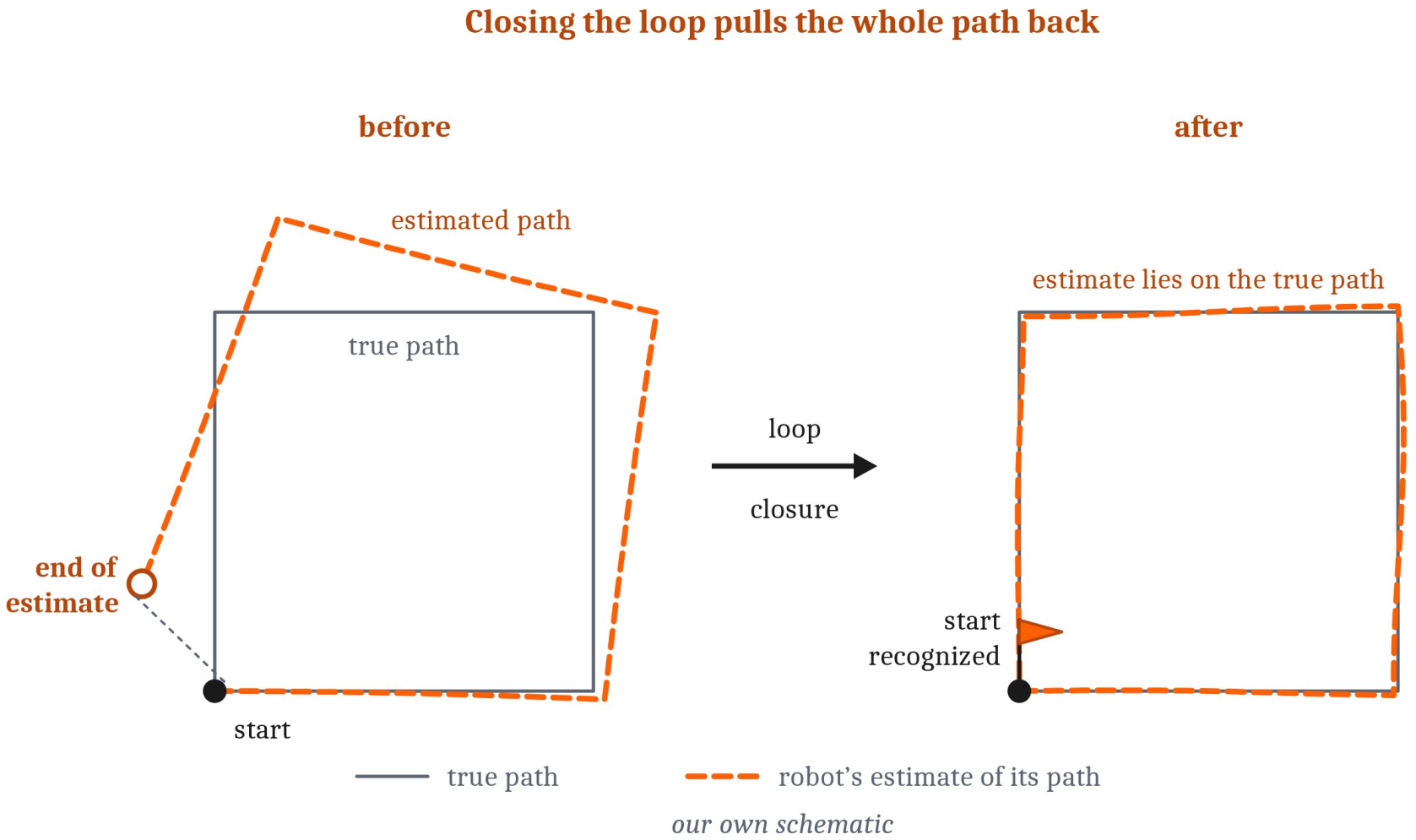
no progress: dropped **no progress: dropped** **progress: worked on**

such as a screen whose channel changes at random (middle)
games chosen freely settled at 70 to 80% correct

Curiosity: a drive to gain information for its own sake.

Machines That Map and Explore

Simultaneous localization and mapping (SLAM): building a map while keeping track of one's position on it. Machines show what a procedure can do, not how brains do it.



The Robot Vacuum, Picked Up

In your group, four minutes.

Many robot vacuums build a map of a home as they clean. Someone picks one up mid-clean and sets it down in another room.

What has gone wrong with its estimate of where it is? Which procedures from this lecture would it need to recover?

Image Sources (1 of 2)

Slide	Image	File	License	Source
3	Searching around enlarged landmarks	fig-snapshot.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
4	Ants on stilts and stumps	fig-stilts.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading, redrawn from the printed values of Wittlinger, Wehner, and Wolf (2006) · own work
5	Drift, and a reset at a landmark	fig-drift-reset.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
6	A ring attractor	fig-ring-attractor.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
7	Converting by adding vectors	fig-vector-sum.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
8	The rotational error	fig-rotational-error.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
9	Repeating codes, combined	fig-repeating-codes.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
10	A predictive map, and choosing a map	fig-using-the-map.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work

Image Sources (2 of 2)

Slide	Image	File	License	Source
11	Choices still ahead	fig-horizon.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading, redrawn from Wilson and colleagues (2014) · own work
11	When to leave a patch	fig-patch-leaving.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
11	Trips out from a home base	fig-home-base-trips.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
12	Two accounts of what curiosity rewards	fig-curiosity.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	Loop closure	fig-loop-closure.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work